

Ulam's Sequence

GREEN-007

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Define Ulam's sequence $1, 2, 3, 4, 6, 8, 11, 13, 16, 18, 26, 28, 36, \dots$ where $u_1 = 1, u_2 = 2$, and u_{n+1} is the smallest number uniquely expressible as $u_i + u_j$ for $i < j \leq n$. Does this sequence have positive density? Can one explain its curious Fourier properties?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>